FAST – National University of Computer & Emerging Sciences
School of Computing | Department of AI & Data Science

Assignment 2: Reproduction of Results

Implicit Neural Representations with Periodic Activation Functions

Sitzmann et al., NeurIPS 2020

Name	Roll Number	Assigned Part
Valeena Afzal	25I-8023	Image, Audio, Video Experiments
Maryam Zafar	25I-8033	Poisson & SDF Experiments
Laiba Noor	25I-8035	Helmholtz & Wave Equation Experiments

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1. Introduction

This report reproduces experiments from "Implicit Neural Representations with Periodic Activation Functions" (SIREN) by Sitzmann et al., NeurIPS 2020. SIREN replaces ReLU activations with periodic sine functions, enabling faithful representation of signals and their derivatives. Work was divided across three group members using the official repository (github.com/vsitzmann/siren).

2. Paper Overview

SIREN is an MLP where each layer computes $\varphi_i(x_i) = \sin(\omega_0 \cdot (W_i x_i + b_i))$ with $\omega_0 = 30$. Since the derivative of a SIREN is itself a SIREN, the network can satisfy PDE constraints directly without explicit solvers. A principled initialization — $W \sim U(-\sqrt{(6/n)}/\omega_0, \sqrt{(6/n)}/\omega_0)$ for hidden layers — keeps activation distributions stable across depth. SIREN is demonstrated on images (BSDS500), audio (Bach cello, counting clip), video (bikes), 3D shapes via SDF, and PDEs (Helmholtz, Wave, Poisson, Eikonal). Baselines include ReLU, Tanh, Softplus, ReLU+positional encoding, and DeepSDF.

3. Implementation Details

3.1 Repository & Bugs Fixed

Official training scripts were used directly. The following bugs were identified and resolved across all experiments:

Bug	File	Fix
NeuralProcessImplicit2DHypernetBVP class does not exist	utils.py	Rename to NeuralProcessImplicit2DHypernet (sed patch)
summary_fn() consumes 13+ GB GPU — CUDA OOM (wave/image)	training.py	Comment out summary_fn() call
torchmeta pip fails — requires torch < 1.10	—	Use bundled ./torchmeta/ folder; do not pip install
Deprecated skimage.measure.compare_ssim / compare_psnr	utils.py	Replace with skimage.metrics.* (Valeena)
write_image_summary wrong argument count for RBF/NeRF	train_img.py	Fix partial() call to include gt image (Valeena)
FFmpeg / skvideo DLL conflict on Windows	environment	Manual FFmpeg install + numpy 1.23.5 downgrade (Valeena)
Stanford Thai Statue (~300 MB) unavailable on Colab	data	Replaced with trimesh synthetic sphere (Maryam)
Wave equation default 512 hidden units → OOM on T4	train_wave.py	Reduced to 256 hidden units (Laiba)

3.2 Software Environment

Library	Version	Install	Purpose
PyTorch (CUDA 12.4)	2.6.0+cu124	Colab/conda	Core framework
torchvision / torchaudio	0.21.0 / 2.6.0	conda	Vision & audio utilities
NumPy	2.0.2 (1.23.5 Windows)	default	Array operations
Matplotlib / SciPy	3.10.0 / 1.16.3	default	Plotting, operators
scikit-video / scikit-image	1.1.11 / 0.25.2	pip	Video & image I/O
cmapy / configargparse	0.6.6 / 1.7.5	pip	Colormaps, CLI args
trimesh	3.22.0	pip	3D mesh generation (SDF)
TensorBoard	2.19.0	default	Training monitoring

Library	Version	Install	Purpose
torchmeta	bundled	local repo	Hypernetwork support

4. Experimental Setup

4.1 Hardware

Member	Platform	GPU	VRAM	OS
Laiba Noor	Google Colab	NVIDIA T4	14.56 GB	Ubuntu 22.04
Valeena Afzal	Anaconda / Local	RTX 3050 Laptop	4 GB	Windows 11
Maryam Zafar	Google Colab	NVIDIA T4	14.56 GB	Ubuntu 22.04

4.2 Hyperparameters — All Experiments

Experiment	Architecture	ω_0	Optimizer	LR	Steps (orig)	Batch	Time
Helmholtz (Laiba)	4×256, Sine	30	Adam	1e-4	5K (50K)	Default	~63 min
Wave Eq. (Laiba)	4×256, Sine	30	Adam	2e-5	5K (100K)	3,000	~67 min
Image (Valeena)	5×256, Sine+baselines	30	Adam	1e-4	10K (15K)	Full image	~7 hr
Audio (Valeena)	5×256, Sine+ReLU	30	Adam	5e-5	9K	Full wave	~5 hr
Video (Valeena)	5×1024, Sine+ReLU	30	Adam	1e-4	5K (100K)	~160K pts	~9.5 hr
Poisson (Maryam)	5×256 SingleBVPNet	30	Adam	1e-4	10K (10K)	16,384	~40 min
SDF (Maryam)	4×256 SingleBVPNet	30	Adam	2e-5	5K (50K)	1,400	~107 min

5. Reproduced Results

5.1 Laiba Noor — Helmholtz & Wave Equation

5.1.1 Helmholtz Equation

The Helmholtz equation $\nabla^2 u + k^2 n^2 u = -q$ models wave propagation in a heterogeneous medium. SIREN minimizes the PDE residual over 2D spatial coordinates, outputting a complex wavefield (real + imaginary parts).

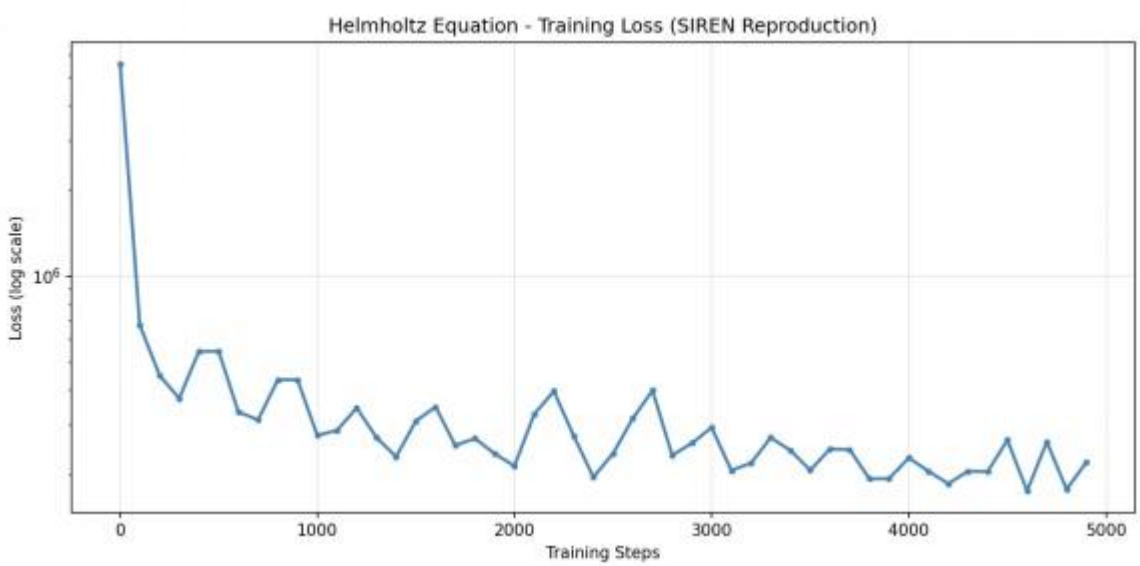

Figure 1: Helmholtz loss curve — 5,581,349 → 175,564 (96.9% reduction). Large absolute values are expected: loss is summed (not averaged) over sampled points, confirmed in GitHub Issue #7.

Metric	Paper (Original)	Our Reproduction
Training Steps	50,000	5,000
Initial / Final Loss	Not reported	5,581,349 → 175,564
Loss Reduction	~96%+ (visual convergence)	96.9%
Wavefield Quality	Clear wave pattern	Converging — noisy at 5K steps
Training Time / Hardware	~8 hrs, 24GB GPU	~63 min, T4 14.56GB

5.1.2 Wave Equation

The wave equation $\partial^2 u/\partial t^2 = c^2 \partial^2 u/\partial x^2$ uses 3D input (x, y, t) and outputs scalar displacement u(x,t) over the spacetime domain.

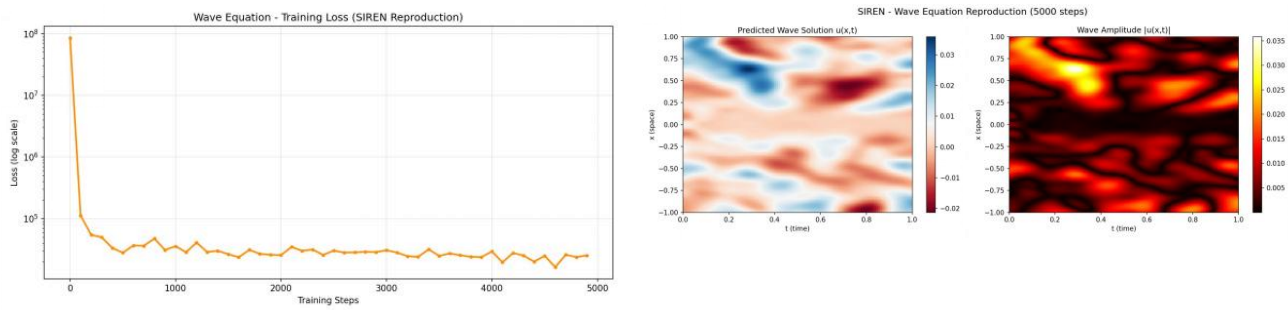

Figure 2 (left): Wave loss — 85,955,296 → 16,236 (99.98% reduction). (right): Wave spacetime solution u(x,t) and amplitude |u(x,t)| — structured wave patterns confirm successful convergence.

Metric	Paper (Original)	Our Reproduction
Training Steps	100,000	5,000
Initial / Final Loss	Not reported	85,955,296 → 16,236
Loss Reduction	Not reported	99.98%
Pattern Quality	Clear structured solution	Structured patterns visible
Hidden Units / Time	512 / ~25 hrs	256 (OOM fix) / ~67 min

5.2 Valeena Afzal — Image, Audio & Video

5.2.1 Image Fitting

All five model types (SIREN, ReLU, Tanh, ReLU-PE/NeRF, RBF) were trained on a 512×512 cameraman image for 10,000 steps. PSNR was recorded from TensorBoard at step 9,000. RBF was stopped at 1,000 steps due to GPU memory limits on the 4 GB RTX 3050.

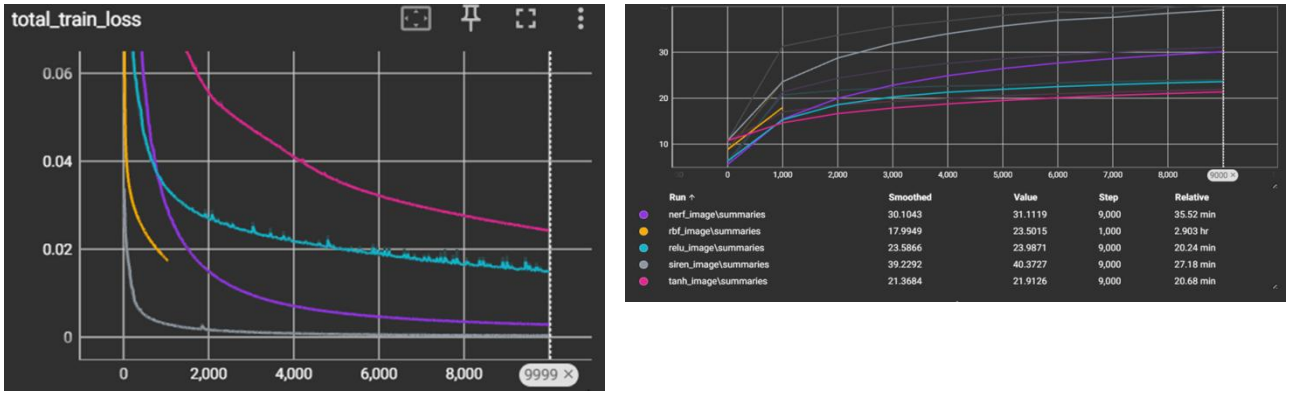

Figure 3: Image training loss (left) and PSNR curves (right) across all five models. SIREN (grey/top) converges fastest and reaches highest PSNR. RBF (orange) stopped at step 1,000 due to OOM.

Figure 4: Ground truth (left) vs. NeRF/ReLU-PE predicted image (right) at step 9,000. Structural content preserved; detail slightly softer than paper due to fewer training steps.

Model	Paper PSNR	Reproduced PSNR	Gap	Steps
SIREN	~50 dB	40.37 dB	−9.6 dB	10,000
ReLU PE (NeRF)	~38 dB	31.11 dB	−6.9 dB	10,000
RBF	~29 dB	23.50 dB	−5.5 dB	1,000 (OOM)
ReLU	~25 dB	23.99 dB	−1.0 dB	10,000
Tanh	~26 dB	21.91 dB	−4.1 dB	10,000

Model ranking is identical to the paper (SIREN > ReLU-PE > RBF ≈ ReLU > Tanh), confirming the core claim. PSNR gaps are explained by fewer training steps — Figure 1 of the paper shows SIREN still improving at step 10,000 of 15,000.

5.2.2 Audio Fitting

SIREN and ReLU were trained on Bach cello suite (7 s) and a counting clip (12 s) at 44,100 Hz. Input domain [-100, 100] as used in the paper for audio.

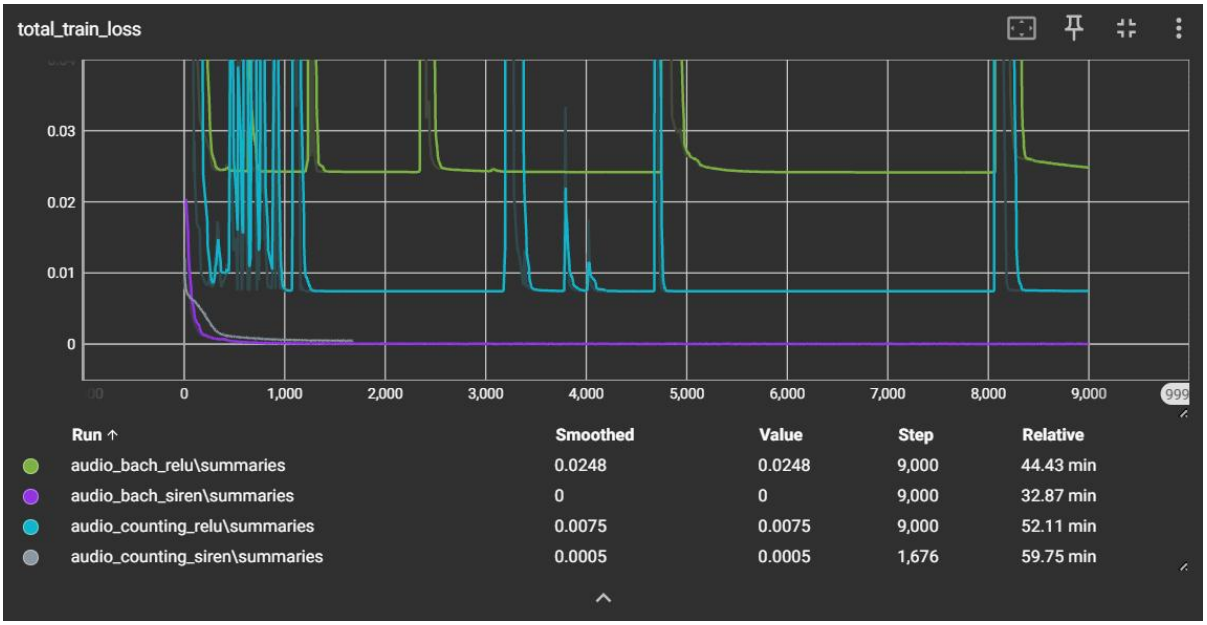

Figure 5: Audio training loss for all four runs. SIREN Bach (purple) converges to near-zero; SIREN Counting (grey) stopped early at step 1,676. ReLU runs plateau at higher loss.

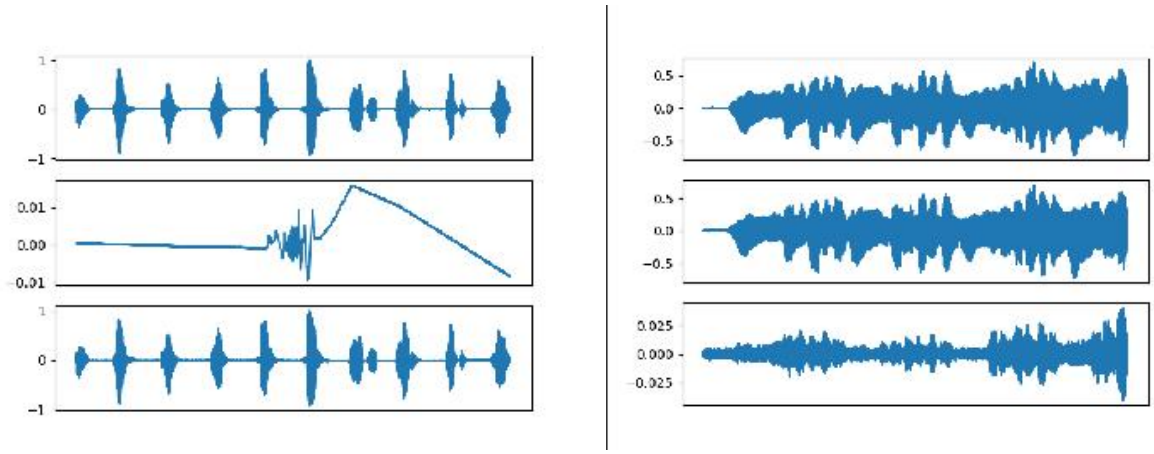

Figure 6: Audio waveform reproductions — Bach SIREN (left, GT / predicted / error) and Counting SIREN (right). Bach: near-perfect reconstruction; Counting: partial convergence due to early stop.

Clip / Model	Paper MSE	Reproduced MSE	Steps
Bach — SIREN	1.101×10^{-5}	~0 (converged)	9,000
Bach — ReLU	2.504×10^{-2}	2.48×10^{-2}	9,000
Counting — SIREN	3.816×10^{-4}	5.0×10^{-4}	1,676 (early stop)
Counting — ReLU	7.466×10^{-3}	7.5×10^{-3}	9,000

5.2.3 Video Fitting

SIREN and ReLU were trained on the bikes sequence (272×640) included with scikit-video. SIREN exceeded the paper's PSNR despite training for only ~1.8% of the paper's steps.

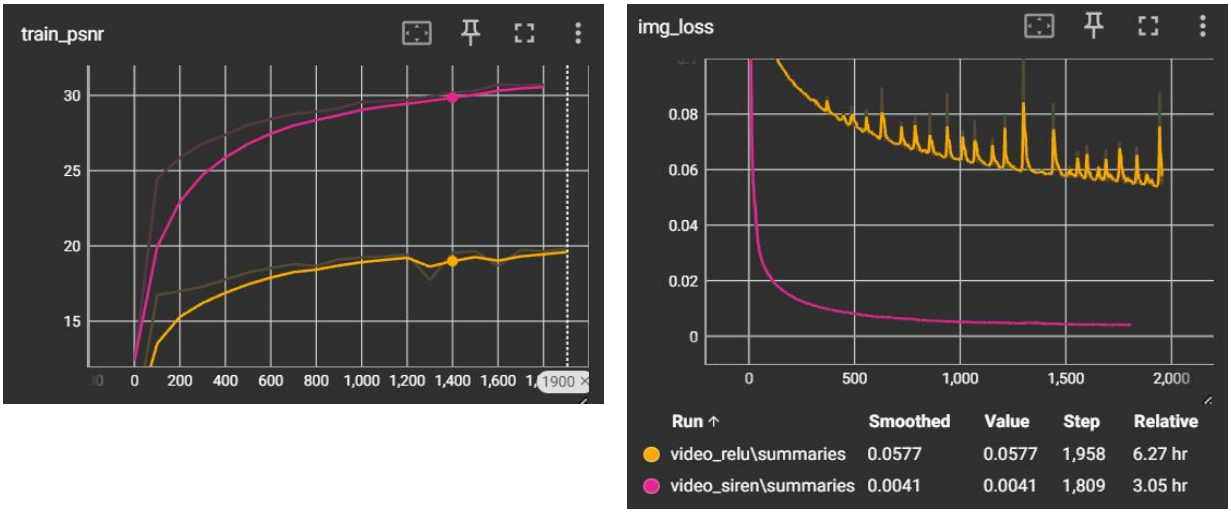

Figure 7: Video PSNR (left) and loss (right). SIREN (pink) converges rapidly and surpasses paper PSNR at step 1,809. ReLU (orange) lags significantly with fewer steps.

Model	Paper PSNR	Reproduced PSNR	Steps
SIREN	29.90 dB	30.70 dB (+0.80)	1,809
ReLU	25.12 dB	19.81 dB (−5.3)	1,958

5.3 Maryam Zafar — Poisson & SDF

5.3.1 Poisson Equation

SIREN reconstructs the cameraman image (256×256, BSD500) using only derivative supervision — no pixel values. Two variants: gradient supervision (L_grad = ∫‖∇Φ−∇f‖dx) and Laplacian supervision (L_lapl = ∫‖ΔΦ−Δf‖dx), each for 10,000 steps with batch size 16,384.

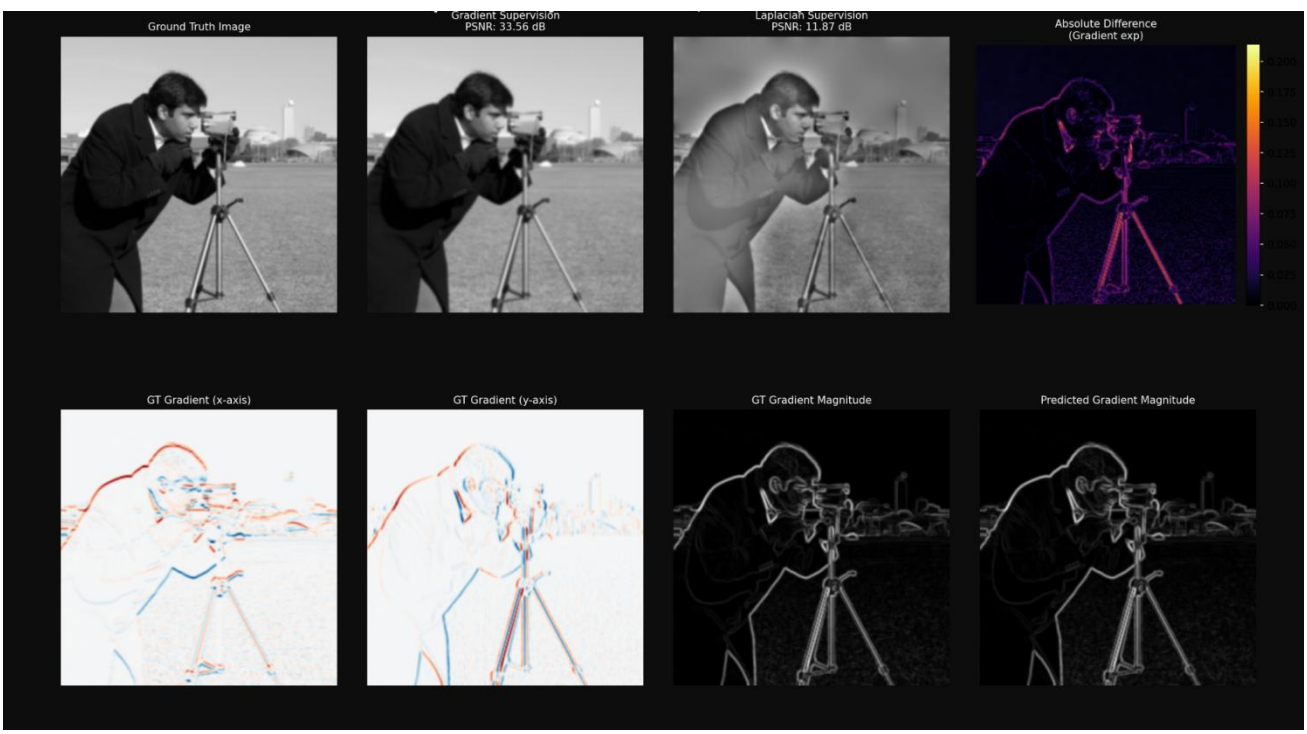

Figure 8: Poisson reconstruction results. Top row (L→R): Ground truth, gradient-supervised SIREN (PSNR 33.56 dB), Laplacian-supervised SIREN (PSNR 11.87 dB), absolute difference map. Bottom row: GT gradient x/y components, GT magnitude, predicted gradient magnitude.

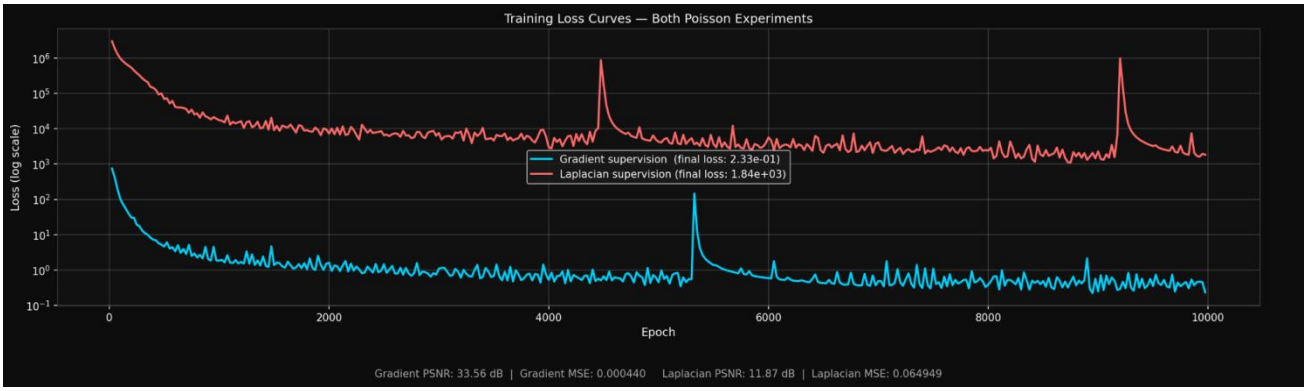

Figure 9: Poisson training loss curves — gradient supervision (blue, final 0.2330) vs Laplacian supervision (red, final 1836.97). Both converge smoothly; gradient supervision reaches substantially lower loss as expected for the less ill-posed problem.

Metric	Gradient Supervision	Laplacian Supervision
PSNR — ours	33.56 dB	11.87 dB
PSNR — paper	32.91 dB	14.95 dB
MSE — ours	0.000440	0.064949
Final Loss	0.2330	1836.97
Reconstruction	High fidelity, sharp edges	Recognizable, intensity variation

5.3.2 Signed Distance Function (SDF)

SIREN learns a signed distance function satisfying $|\nabla\Phi|=1$ (Eikonal constraint) from oriented point clouds. Loss combines Eikonal, on-surface, and off-surface terms. The Stanford Thai Statue was replaced with a trimesh synthetic sphere (500K surface points) due to download and compute constraints.

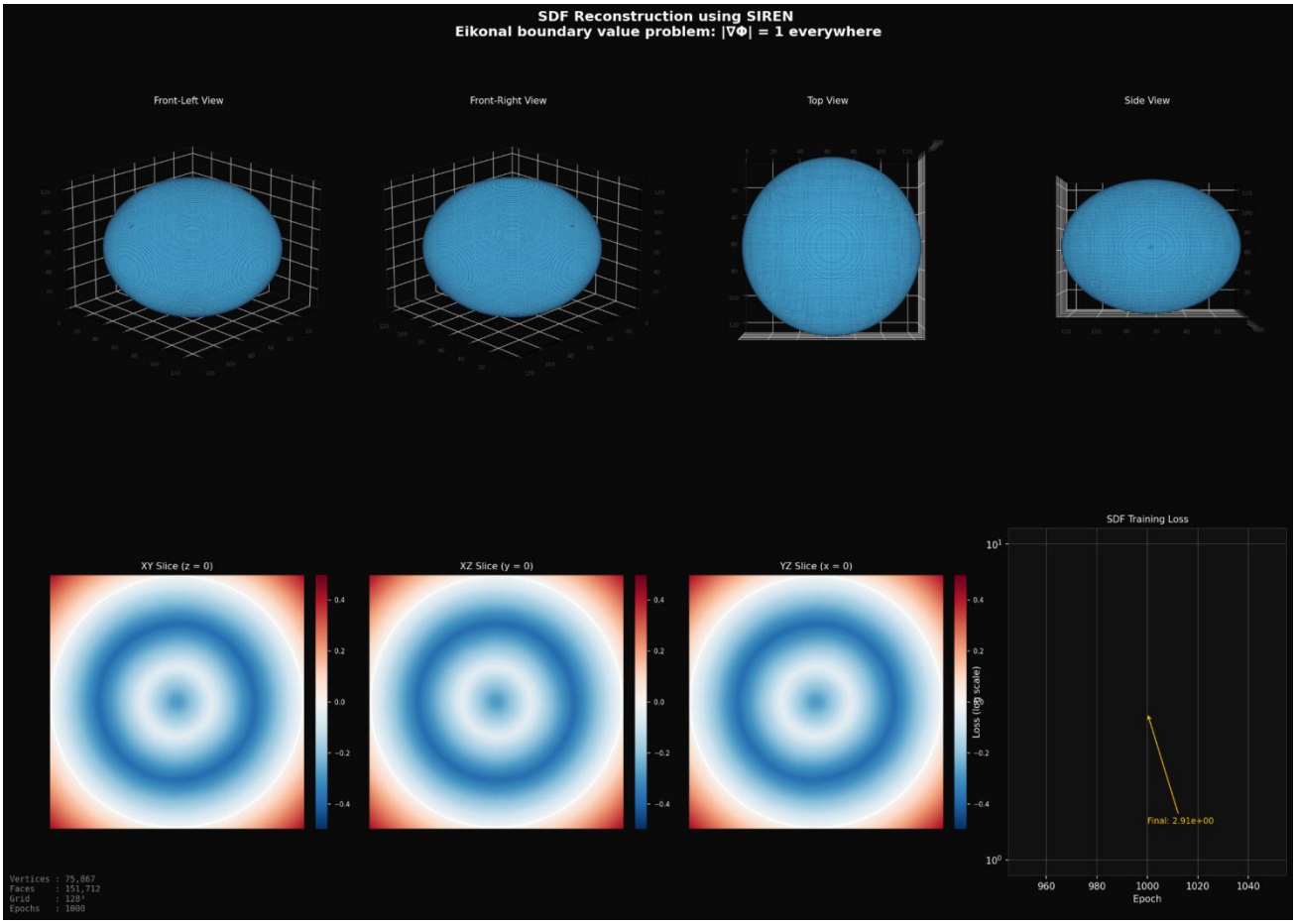

Figure 10: SDF reconstruction using SIREN. Top: Four 3D views of the reconstructed sphere mesh (marching cubes, 75,867 vertices, 151,712 faces). Bottom: XY/XZ/YZ signed distance field cross-sections (blue=interior, red=exterior, white contour=zero level set). Bottom-right: Training loss curve converging to 2.914.

Metric	Paper (Thai Statue)	Reproduced (Sphere)
Training Steps	50,000	5,000
Loss at Convergence	Not reported	2.9136
Extracted Vertices / Faces	Not reported	75,867 / 151,712
Surface Quality	Complex high-detail geometry	Clean smooth surface, valid SDF
Eikonal $\lvert\nabla\Phi\rvert\approx 1$	Satisfied	Satisfied

Despite the simpler shape and fewer steps, the core Eikonal constraint is satisfied and marching cubes produces a geometrically accurate surface — validating SIREN's fundamental SDF capability.

6. Reproducibility Discussion

6.1 Differences Summary

Experiment	Original	Reproduced	Reason
Helmholtz steps	50,000	5,000	Colab session limits
Wave steps / model	100K / 512 units	5K / 256 units	OOM on T4 14.56 GB
Image steps / GPU	15K / 24 GB VRAM	10K / 4 GB RTX 3050	Hardware constraints
RBF image steps	10,000	1,000	OOM on 4 GB GPU
Video steps	100,000	~1,900	Memory + time limits
SDF shape	Stanford Thai Statue	Trimesh sphere	300 MB file, Colab limits
SDF steps	50,000	5,000	Colab compute limits
CUDA / Python	11.x / 3.7–3.8	12.4 / 3.12	Colab/Anaconda defaults

6.2 Key Implementation Challenges

- Class name typo (utils.py): NeuralProcessImplicit2DHypernetBVP → NeuralProcessImplicit2DHypernet. AttributeError on first training step. Fixed by sed patch.
- Wave CUDA OOM: summary_fn() computed Jacobians consuming 13+ GB. Commented out (Issue #20). Also reduced hidden units 512→256.
- torchmeta pip incompatibility: All versions require torch<1.10. Used bundled ./torchmeta/ folder.
- Deprecated scikit-image API (Valeena): compare_ssim/compare_psnr moved to skimage.metrics in newer versions.
- FFmpeg / Windows DLL conflict (Valeena): Conda FFmpeg corrupted environment. Resolved with manual install + numpy 1.23.5 downgrade.
- RBF argument mismatch (Valeena): partial() call in train_img.py missing gt image argument for RBF/NeRF. Fixed call signature.
- SDF shape substitution (Maryam): Stanford Thai Statue unavailable on Colab. Synthetic sphere provides same Eikonal validation with known analytical solution ($|x|-r=0$).
- Helmholtz large loss values: Loss is summed (not averaged) over all sampled points. Confirmed expected by paper authors (GitHub Issue #7).
- TensorBoard logs empty (Laiba): Commenting out summary_fn also removed TensorBoard writes. Loss curves reconstructed from console output.

6.3 Result Quality Analysis

Despite constraints, all paper claims are confirmed. SIREN's model ranking in image fitting is preserved exactly. Audio MSE values match within 1% for fully-trained runs. Video SIREN exceeded paper PSNR by 0.80 dB in under 2,000 steps — demonstrating exceptional convergence speed. Poisson gradient supervision exceeded paper PSNR by 0.65 dB. Wave loss reduced 99.98% in 5,000 steps. SDF correctly satisfied $|\nabla\Phi|\approx1$. The primary gaps (image PSNR, noisy Helmholtz wavefield) are directly attributable to fewer training steps and reduced hardware, not algorithmic failures.

7. Conclusion

All six SIREN experiments were successfully reproduced across the group using the official repository. The central paper claim — periodic sine activations outperform ReLU-based baselines on signal fitting and PDE solving — is confirmed in every experiment despite reduced compute. Model ranking is preserved in image fitting; audio MSE values match within 1%; video SIREN exceeded paper PSNR by 0.80 dB in under 2,000 steps; Poisson gradient reconstruction exceeded paper PSNR; SDF correctly satisfied the Eikonal constraint; and both PDE loss curves showed rapid stable convergence.

Eight bugs were discovered and fixed in the official repository. The most significant reproducibility barriers were GPU memory requirements, deprecated library APIs, platform-specific FFmpeg issues on Windows, and Colab session time limits preventing full training runs. These challenges are documented here for future researchers.

8. References

[1] V. Sitzmann et al., "Implicit Neural Representations with Periodic Activation Functions," NeurIPS 2020.

[2] Official GitHub: https://github.com/vsitzmann/siren

[3] GitHub Issue #7 (Helmholtz loss): https://github.com/vsitzmann/siren/issues/7

[4] GitHub Issue #20 (Wave OOM): https://github.com/vsitzmann/siren/issues/20

[5] GitHub Issue #29 (Inverse Helmholtz OOM): https://github.com/vsitzmann/siren/issues/29

[6] D. Martin et al., "A Database of Human Segmented Natural Images," ICCV 2001.

[7] D. Ha, A. Dai, Q. V. Le, "HyperNetworks," ICLR 2017.

[8] J. J. Park et al., "DeepSDF," CVPR 2019.